

Report Details

Project Title

Housing Data Analysis using Tableau

Internship Domain

Data Analytics

Team ID

SWTID-2026-3991

SkillWallet ID

SWUID2026233897

1. INTRODUCTION

1.1 Project Overview

In today's digital era, the real estate industry generates a massive amount of housing-related data every day. This data contains valuable information about property prices, house characteristics, location, renovation history, and other factors that influence the housing market. However, analyzing large datasets using traditional methods such as spreadsheets is time-consuming and often fails to reveal meaningful patterns. Data visualization tools help transform raw data into interactive dashboards that simplify analysis and support better decision-making.

The project "**Housing Data Analysis using Tableau**" was developed as part of the Data Analytics Internship to analyze housing sales data and present meaningful insights through interactive visualizations. Tableau Desktop was used as the primary Business Intelligence (BI) tool because of its powerful data visualization capabilities and user-friendly interface. The project focuses on identifying relationships between different housing attributes such as sale price, house age, basement area, living area, number of bedrooms, bathrooms, floors, and renovation status.

The development process began with collecting and understanding the housing dataset. The dataset was carefully examined for missing values, duplicate records, and inconsistencies. After preprocessing and cleaning the data, it was imported into Tableau Desktop, where multiple worksheets, dashboards, and story points were created. Various charts, graphs, and Key Performance Indicators (KPIs) were designed to present the information in an interactive and visually appealing manner.

The dashboards allow users to filter data dynamically and analyze housing trends from different perspectives. For example, users can compare house prices based on renovation status, evaluate the impact of basement area on property value, study house age distribution, and identify factors affecting residential property prices. Interactive filters and visualizations enable users to explore the dataset without requiring advanced technical knowledge.

This project demonstrates how Business Intelligence tools can convert raw housing data into valuable business insights. The developed dashboard assists real estate analysts, property investors, home buyers, and researchers in understanding market trends and making informed decisions based on data. By combining

data preprocessing, visualization, and interactive analytics, the project provides an efficient solution for exploring housing market information.

1.2 Purpose

The primary purpose of this project is to develop an interactive data visualization dashboard that enables efficient analysis of housing sales data using Tableau Desktop. The project aims to simplify the interpretation of complex datasets by converting numerical information into meaningful charts, graphs, and dashboards that support data-driven decision-making.

The specific objectives of the project are:

- To analyze housing sales data using modern Business Intelligence techniques.
- To clean, preprocess, and organize the dataset before visualization.
- To identify the factors influencing house prices, including renovation status, basement area, house age, living area, and other property characteristics.
- To develop interactive dashboards that allow users to filter and explore housing data effectively.
- To present important business insights using KPIs, charts, maps, and story points.
- To improve the understanding of housing market trends through visual analytics.
- To demonstrate the practical application of Tableau Desktop in solving real-world data analytics problems.
- To support informed decision-making for property buyers, sellers, real estate analysts, and business professionals.

The project also aims to enhance practical knowledge of data analytics by applying concepts such as data preprocessing, exploratory data analysis, dashboard development, and business intelligence reporting. It serves as an example of how data visualization can transform raw data into actionable insights that benefit both businesses and end users.

2. IDEATION PHASE

2.1 Problem Statement

The real estate industry produces large volumes of housing data containing information about property prices, house size, number of bedrooms, bathrooms, renovation status, basement area, geographical location, and many other attributes. Although this data is valuable, it is often stored in spreadsheets or databases where identifying meaningful trends is difficult. Manual analysis is time-consuming and does not provide quick business insights.

Property buyers, investors, and real estate analysts frequently need to compare houses based on different factors before making decisions. Without proper visualization tools, it becomes challenging to understand how different variables influence housing prices. Decision-makers may overlook important trends that could affect investment or purchasing decisions.

To address this challenge, this project proposes the development of an interactive Tableau dashboard that transforms raw housing data into meaningful visualizations. The dashboard allows users to analyze house prices, compare property features, evaluate renovation effects, study house age, and examine basement area using interactive charts, KPIs, and filters. This approach improves decision-making by presenting complex information in a simple, visual, and user-friendly manner.

2.2 Empathy Map Canvas

The Empathy Map Canvas helps identify the needs, expectations, and challenges of the intended users of the Housing Data Analysis Dashboard.

User

Real Estate Analyst, Property Investor, Home Buyer, Business Analyst

Says

- "I need accurate information before investing in a property."
- "I want to compare house prices quickly."
- "The data should be easy to understand."
- "I need reliable insights rather than large spreadsheets."

Thinks

- Which property offers the best value for money?
- Does renovation significantly increase the selling price?
- Which house characteristics have the greatest impact on price?
- How can I identify market trends efficiently?

Does

- Compares multiple properties.
- Studies market reports.
- Reviews housing data.
- Uses dashboards and reports for decision-making.
- Evaluates investment opportunities.

Feels

- Confused when dealing with large datasets.
- Concerned about making incorrect investment decisions.
- Frustrated by complicated spreadsheets.
- Confident when clear visual insights are available.

Pain Points

- Large amounts of raw housing data.
- Difficulty identifying important trends.
- Manual data analysis consumes time.
- Limited visualization in traditional spreadsheets.

Needs

- Interactive dashboards.
 - Accurate housing analysis.
 - Easy-to-understand visual reports.
 - Quick access to important business insights.
 - Reliable support for property-related decisions.
-

2.3 Brainstorming

Before developing the dashboard, several ideas were generated to determine the most effective way to analyze the housing dataset. The brainstorming process focused on identifying features that would provide meaningful insights while keeping the dashboard simple, interactive, and easy to use.

The following ideas were considered during the brainstorming stage:

- Develop an interactive Tableau dashboard.
- Display important KPIs such as Average Sale Price and Basement Area.
- Analyze the relationship between house age and selling price.
- Compare renovated and non-renovated houses.
- Study the impact of basement area on house value.
- Analyze bedrooms, bathrooms, and floors.
- Create visualizations showing the distribution of sale prices.
- Use filters to allow users to explore the dataset dynamically.
- Organize multiple dashboards into Tableau Story Points.
- Generate business insights to support decision-making.

After evaluating these ideas, the project selected the interactive dashboard approach because it provides a balance between simplicity, usability, and analytical capability. Tableau was chosen as the primary visualization tool due to its ability to create interactive charts, dashboards, and stories without requiring complex programming.

The brainstorming phase established the foundation for the remaining stages of the project, including requirement analysis, dashboard design, implementation, testing, and final deployment.

3. REQUIREMENT ANALYSIS

Requirement analysis is one of the most important stages in the software development life cycle. During this phase, the objectives of the project, user expectations, system requirements, and technological resources were carefully identified and documented. A proper requirement analysis ensures that the developed solution meets user needs and functions efficiently.

For the **Housing Data Analysis using Tableau** project, the requirement analysis focused on understanding the needs of users who wish to analyze housing market data quickly and accurately through interactive dashboards. The analysis also identified the necessary software, hardware, dataset, and visualization techniques required for successful project implementation.

3.1 Customer Journey Map

The Customer Journey Map describes the sequence of actions performed by users while interacting with the Housing Data Analysis Dashboard. It also highlights user goals, experiences, and expected outcomes at each stage.

Stage	User Activity	User Experience
Awareness	User wants to understand housing market trends.	Needs reliable and easy-to-understand information.
Data Exploration	Opens the Tableau dashboard and explores available visualizations.	Finds dashboards interactive and informative.

Stage	User Activity	User Experience
Analysis	Applies filters such as renovation status, house age, or basement area to compare properties.	Gains meaningful insights without manually processing data.
Decision Making	Uses dashboard results to compare different housing characteristics.	Makes informed property-related decisions with greater confidence.
Review	Revisits dashboards whenever additional analysis is required.	Saves time through reusable visual reports and interactive filters.

The customer journey demonstrates how an interactive dashboard simplifies the analysis process compared to traditional spreadsheet-based methods.

3.2 Solution Requirement

The proposed solution is an interactive Tableau dashboard that transforms raw housing data into meaningful visual reports and business insights. The dashboard enables users to analyze multiple housing characteristics simultaneously using dynamic filters, KPIs, charts, and story points.

Functional Requirements

The system should:

- Import the housing dataset successfully.
- Display important Key Performance Indicators (KPIs).
- Generate interactive charts and graphs.
- Provide filtering options for users.
- Analyze house prices based on multiple factors.
- Present dashboards and story points.
- Allow easy interpretation of housing market trends.
- Support business decision-making through visual analytics.

Non-Functional Requirements

The system should:

- Be easy to use for both technical and non-technical users.
- Generate dashboards with acceptable performance.
- Maintain data accuracy and consistency.
- Present information clearly using visual elements.
- Be reliable and easy to maintain.
- Produce professional-quality reports and dashboards.

3.3 Data Flow Diagram

The project follows a simple and systematic data flow from data collection to final visualization.

Housing Dataset (CSV File)



Data Cleaning & Validation



Data Preprocessing



Tableau Desktop Import



Dashboard & KPI Development



Interactive Visualizations



Business Insights & Reports

Description

The housing dataset is first collected in CSV format. During preprocessing, missing values, duplicate records, and inconsistencies are removed to improve data quality. The cleaned dataset is then imported into Tableau Desktop, where calculated fields, worksheets, dashboards, and story points are created. Finally, users interact with dashboards to obtain meaningful business insights.

3.4 Technology Stack

The project utilizes modern Business Intelligence tools for data analysis and visualization.

Technology	Purpose
Tableau Desktop	Creating interactive dashboards and visualizations
CSV Dataset	Source of housing data
Microsoft Excel	Initial data inspection and formatting
GitHub	Version control and project repository
Windows Operating System	Development environment

Software Requirements

- Tableau Desktop
- Microsoft Excel

- GitHub
- Web Browser
- PDF Reader

Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB (8 GB Recommended)
- Storage: Minimum 10 GB Free Space
- Display Resolution: 1366 × 768 or higher

Summary

The Requirement Analysis phase identified the functional and non-functional requirements of the Housing Data Analysis project. It defined the customer journey, described the system workflow through the Data Flow Diagram, and listed the technologies required for development. These requirements served as the foundation for designing and implementing the interactive Tableau dashboard.

4. PROJECT DESIGN

Project design is the process of transforming the identified requirements into a structured solution that can be implemented efficiently. It provides a clear understanding of how the system will function, how data will be processed, and how users will interact with the final application. A well-designed system improves usability, simplifies maintenance, and ensures that the project objectives are successfully achieved.

For the **Housing Data Analysis using Tableau** project, the design focuses on transforming raw housing data into meaningful visual insights through interactive dashboards. The system is designed to provide users with an easy-to-use interface where they can explore housing data, apply filters, compare property characteristics, and obtain valuable business insights without requiring technical expertise.

4.1 Problem–Solution Fit

The real estate sector generates a large amount of housing data containing information such as sale price, number of bedrooms, bathrooms, basement area, living area, house age, renovation status, and geographical location. Although this information is valuable, it is difficult to analyze manually using spreadsheets because large datasets are time-consuming to interpret and often fail to reveal hidden trends.

The proposed solution addresses these challenges by developing an interactive Tableau dashboard that converts raw numerical data into graphical visualizations. Instead of manually comparing hundreds or thousands of records, users can easily identify trends through charts, KPIs, filters, and dashboards.

The solution is well suited to the identified problem because it provides:

- Interactive data visualization instead of static reports.
- Faster analysis of large housing datasets.
- Better understanding of property price trends.
- Easy comparison of different housing characteristics.
- Improved decision-making using graphical insights.
- User-friendly dashboards requiring minimal technical knowledge.

By replacing manual analysis with interactive business intelligence dashboards, the project enables users to explore data efficiently and make informed decisions based on reliable visual evidence.

4.2 Proposed Solution

The proposed solution is an interactive Business Intelligence system developed using Tableau Desktop. The system imports the cleaned housing dataset and transforms it into multiple worksheets, dashboards, and story points that present important housing market information in a simple and visually attractive format.

The development process begins with collecting and preprocessing the housing dataset. During preprocessing, missing values, duplicate records, and inconsistencies are removed to improve data quality. The cleaned dataset is then imported into Tableau Desktop, where various visualizations are created using charts, KPIs, filters, and calculated fields.

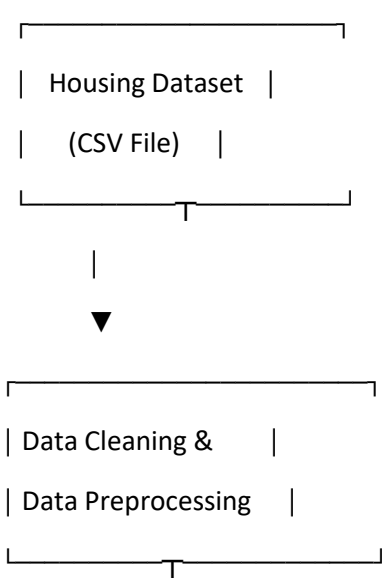
The dashboard provides users with the ability to:

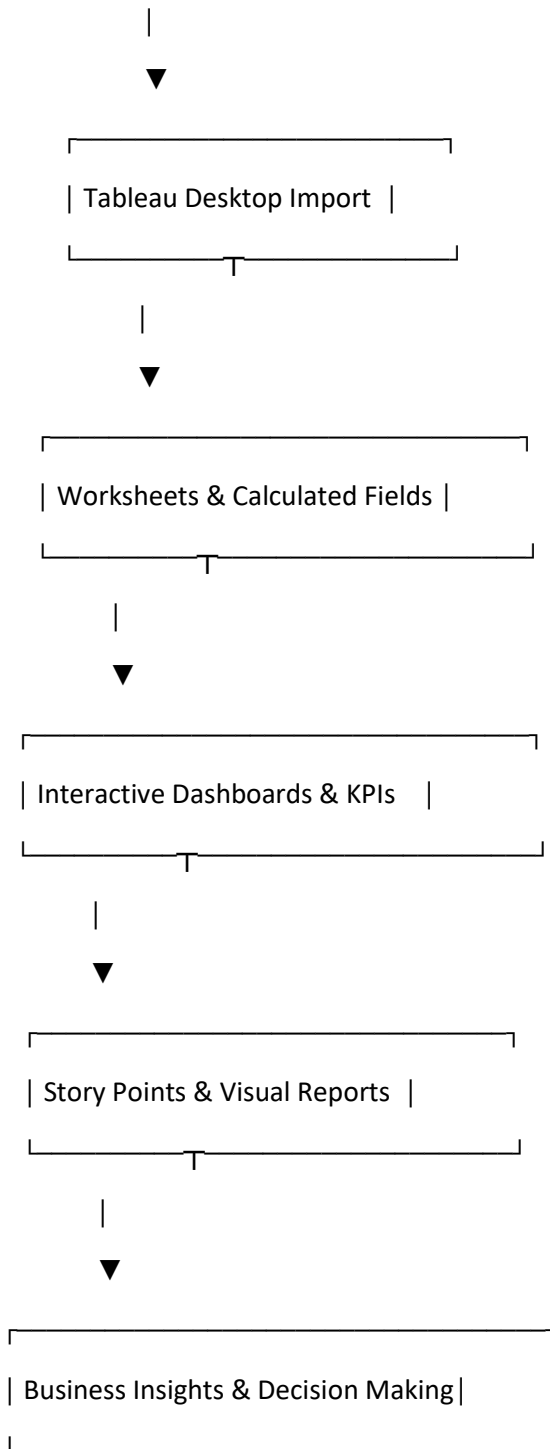
- View important housing KPIs.
- Compare sale prices of different properties.
- Analyze the effect of renovation on property value.
- Study the relationship between basement area and sale price.
- Examine house age distribution.
- Filter data dynamically based on different attributes.
- Navigate through multiple dashboards using story points.

The proposed solution reduces the complexity of housing data analysis while providing accurate and meaningful business insights. It supports property buyers, investors, researchers, and real estate analysts by presenting information in an interactive and easy-to-understand manner.

4.3 Solution Architecture

The solution architecture illustrates the complete workflow of the Housing Data Analysis project, starting from raw data collection and ending with business decision-making through interactive dashboards.





Architecture Description

The architecture begins with the collection of the housing dataset in CSV format. The dataset undergoes preprocessing to eliminate missing values, duplicate entries, and inconsistent data. Once cleaned, the data is imported into Tableau Desktop, where worksheets and calculated fields are created to perform detailed analysis.

These worksheets are combined into interactive dashboards containing charts, KPIs, maps, and filters. Story points are then used to organize the dashboards into a logical sequence, allowing users to understand the analysis step by step. Finally, the dashboard provides meaningful business insights that support informed decision-making in the real estate sector.

The Project Design phase converts the identified requirements into a structured solution. The proposed Tableau-based architecture provides an efficient and interactive platform for housing data analysis by integrating data preprocessing, visualization, dashboard development, and business intelligence reporting. The design ensures that users can explore housing data easily, identify important trends, and make informed decisions based on visual analytics.

5. PROJECT PLANNING & SCHEDULING

Project planning is an essential phase in the Software Development Life Cycle (SDLC). It provides a structured approach for organizing project activities, allocating resources, estimating timelines, and monitoring progress. Proper planning ensures that the project is completed systematically within the given duration while maintaining the desired quality.

For the **Housing Data Analysis using Tableau** project, the planning phase involved dividing the project into manageable stages, beginning with understanding the problem statement and ending with the final deployment of dashboards and documentation. Each phase was completed in a logical sequence to ensure smooth development and successful implementation.

5.1 Project Planning

The Housing Data Analysis project was planned according to the different phases of the internship. Each phase contributed to the successful completion of the final dashboard and documentation.

Phase 1: Project Selection and Ideation

The first step was selecting an appropriate project for the Data Analytics internship. After exploring different domains, the Housing Data Analysis project was chosen because it provides an opportunity to apply Business Intelligence techniques for solving real-world problems. During this phase, the problem statement, empathy map, and brainstorming activities were completed to understand the requirements of potential users.

Phase 2: Requirement Analysis

After identifying the project objectives, the system requirements were analyzed. Functional requirements, non-functional requirements, customer journey mapping, data flow diagrams, and the technology stack were prepared. This phase ensured that all necessary resources and project objectives were clearly defined before development began.

Phase 3: Dataset Collection and Preparation

The housing dataset was collected in CSV format and carefully examined for quality. The preprocessing stage included:

- Identifying missing values.

- Removing duplicate records.
- Validating numerical data.
- Organizing fields into appropriate formats.
- Preparing the dataset for visualization.

A clean dataset improves the reliability and accuracy of dashboard analysis.

Phase 4: Dashboard Development

After preprocessing, the dataset was imported into Tableau Desktop. Multiple worksheets were created to analyze different housing attributes. Interactive dashboards were then developed by combining charts, KPIs, filters, and calculated fields.

The dashboards focused on:

- Average Sale Price
- Basement Area Analysis
- House Age Analysis
- Renovation Status
- Sale Price Distribution
- Property Characteristics
- Interactive Filters

Story Points were also created to organize the dashboards into a logical presentation.

Phase 5: Testing and Validation

Once dashboard development was completed, each worksheet and dashboard was tested to ensure correct functionality. Data values displayed in Tableau were verified against the original dataset to confirm accuracy. Filters, charts, and KPIs were also tested to ensure they responded correctly to user interactions.

Phase 6: Documentation and Deployment

The final phase involved preparing the internship documentation, GitHub repository, project report, and demonstration materials. All project files were organized according to the internship submission guidelines. The completed dashboards and supporting documents were uploaded to GitHub for version control and future reference.

Project Schedule

Project Phase	Major Activities
Ideation	Problem identification, empathy map, brainstorming
Requirement Analysis	Customer journey, solution requirements, DFD, technology stack
Project Design	Architecture design, solution planning
Data Preparation	Data cleaning, preprocessing, validation

Project Phase	Major Activities
Dashboard Development	Worksheets, KPIs, charts, dashboards, story points
Testing	Dashboard validation, KPI verification, filter testing
Documentation	Final report, GitHub repository, project submission

Project Milestones

The following milestones were achieved during the project:

- ✓ Project topic finalized.
- ✓ Housing dataset collected.
- ✓ Dataset cleaned and preprocessed.
- ✓ Tableau workbook created.
- ✓ Worksheets designed.
- ✓ Interactive dashboards developed.
- ✓ Story Points completed.
- ✓ Dashboard testing completed.
- ✓ GitHub repository created.
- ✓ Internship documentation prepared.
- ✓ Final project report completed.

Risk Management

During project development, several challenges were identified and addressed to ensure successful completion.

Potential Risk	Mitigation Strategy
Missing or inconsistent data	Data preprocessing and validation before visualization
Incorrect dashboard calculations	Verification of KPIs and calculated fields
Performance issues	Optimized worksheets and minimized unnecessary calculations
Documentation errors	Regular review and cross-checking with internship guidelines
Data interpretation mistakes	Validation using multiple visualizations and business insights

Summary

The Project Planning and Scheduling phase provided a structured roadmap for completing the Housing Data Analysis project. By dividing the work into clearly defined stages, the project progressed systematically from ideation to deployment. Careful planning ensured timely completion of dashboard development, testing, documentation, and final submission while maintaining the overall quality of the project.

6. FUNCTIONAL AND PERFORMANCE TESTING

Testing is a critical phase in the software development process that ensures the developed system performs as expected and meets user requirements. It helps identify errors, verify functionality, and improve the overall quality of the project. For the **Housing Data Analysis using Tableau** project, testing was carried out after the completion of dashboard development to verify the correctness of data visualizations, dashboard interactions, and overall system performance.

The testing process focused on validating the accuracy of the imported dataset, checking the functionality of interactive dashboards, verifying calculated fields and KPIs, and ensuring that filters and charts responded correctly to user interactions. The performance of the dashboard was also evaluated to confirm that it loaded efficiently and provided smooth navigation.

6.1 Performance Testing

Testing Objectives

The primary objectives of testing were:

- To verify that the housing dataset was imported correctly.
- To ensure that all worksheets displayed accurate information.
- To validate the correctness of KPIs and calculated fields.
- To test dashboard filters and interactive controls.
- To verify that dashboards responded smoothly to user actions.
- To ensure that visualizations accurately represented the underlying data.
- To confirm that the dashboard provided reliable business insights.

Functional Testing

Functional testing was performed to ensure that every feature of the Tableau dashboard worked according to the project requirements.

Test Case ID	Feature Tested	Expected Result	Actual Result	Status
TC-01	Dataset Import	Dataset imported successfully	Imported successfully	Pass
TC-02	Worksheet Creation	Worksheets display correct data	Correctly displayed	Pass
TC-03	Dashboard Loading	Dashboard opens without errors	Loaded successfully	Pass
TC-04	KPI Display	KPIs show correct values	Displayed accurately	Pass
TC-05	Interactive Filters	Filters update all charts correctly	Working properly	Pass
TC-06	Story Points	Story navigation functions correctly	Working properly	Pass
TC-07	Charts and Graphs	Charts represent data accurately	Verified	Pass
TC-08	User Interaction	Dashboard responds smoothly	Verified	Pass

Performance Testing

Performance testing was conducted to evaluate the efficiency and responsiveness of the Tableau dashboard.

Parameter	Observation
Dashboard Loading Time	Fast and responsive
Data Visualization	Accurate and smooth
Filter Response	Immediate response
Dashboard Navigation	Smooth
KPI Calculation	Accurate
Story Navigation	Successful
Overall Performance	Satisfactory

The dashboard demonstrated stable performance during testing. No significant delays or system failures were observed while interacting with the visualizations.

Data Validation

To ensure the reliability of the dashboard, the displayed values were compared with the original housing dataset.

The following validation activities were performed:

- Verified numerical values against the source dataset.
- Confirmed the correctness of calculated fields.
- Cross-checked KPIs with Tableau calculations.
- Validated dashboard filters using multiple combinations.
- Ensured charts represented the dataset accurately.

No inconsistencies were identified during validation, indicating that the dashboard correctly represented the housing data.

User Acceptance Testing (UAT)

User Acceptance Testing was carried out by interacting with the completed dashboard from the perspective of an end user. The dashboard was evaluated based on usability, clarity, and ease of navigation.

The following aspects were assessed:

- Ease of understanding the dashboard.
- Simplicity of applying filters.
- Clarity of charts and visualizations.
- Readability of KPIs.
- Overall user experience.

The dashboard was found to be intuitive, interactive, and suitable for analyzing housing market data.

Testing Summary

The testing phase confirmed that all major components of the Housing Data Analysis project functioned correctly. The Tableau dashboards successfully displayed accurate housing information, interactive filters operated as expected, and KPIs generated reliable results. Performance testing demonstrated that the dashboards responded efficiently without noticeable delays, providing a smooth user experience.

The successful completion of testing indicates that the developed system satisfies the functional and performance requirements defined during the requirement analysis phase.

Table 6.1: Overall Testing Result

Testing Category	Status
Dataset Validation	✓ Passed
Dashboard Functionality	✓ Passed
KPI Verification	✓ Passed
Filter Testing	✓ Passed
Story Navigation	✓ Passed
Visualization Accuracy	✓ Passed
Performance Testing	✓ Passed
User Acceptance Testing	✓ Passed

Chapter Summary

The Functional and Performance Testing phase verified that the Housing Data Analysis dashboard performs accurately, efficiently, and reliably. All dashboard components were tested successfully, ensuring that users can interact with the visualizations confidently and obtain meaningful insights from the housing dataset.

7. RESULTS

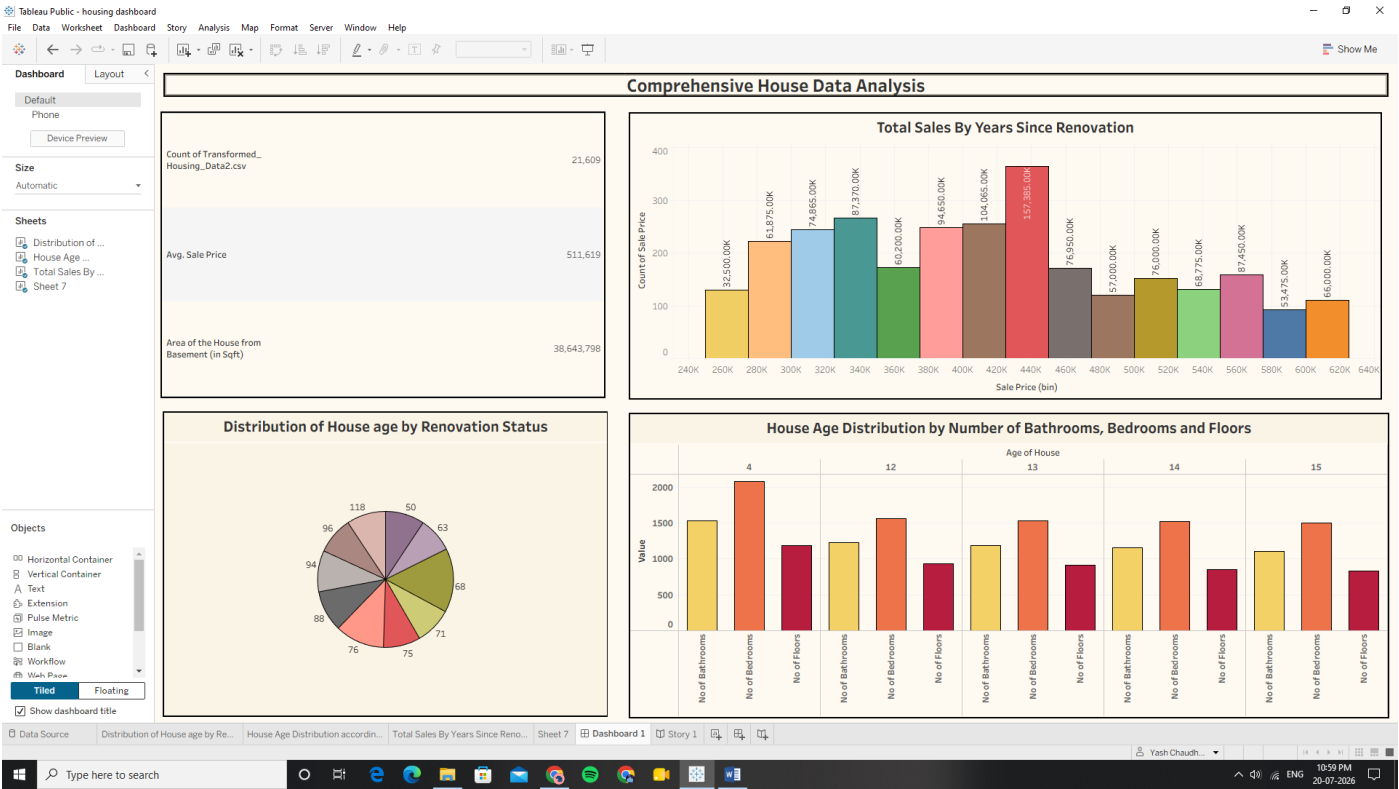
The final phase of the project involved analyzing the housing dataset through interactive Tableau dashboards and interpreting the generated visualizations. The developed dashboards successfully transformed raw housing data into meaningful graphical representations, enabling users to understand market trends, compare property characteristics, and make informed decisions.

The results obtained from the project demonstrate that Tableau is an effective Business Intelligence tool for exploring large datasets and presenting complex information in an interactive and user-friendly format. The dashboards provide valuable insights into housing prices, renovation status, house age, basement area, and other important housing attributes.

The project successfully achieved its objective of converting raw housing data into visually appealing dashboards that support data-driven decision-making.

7.1 Output Screenshots

Figure 7.1: Housing Data Analysis Dashboard

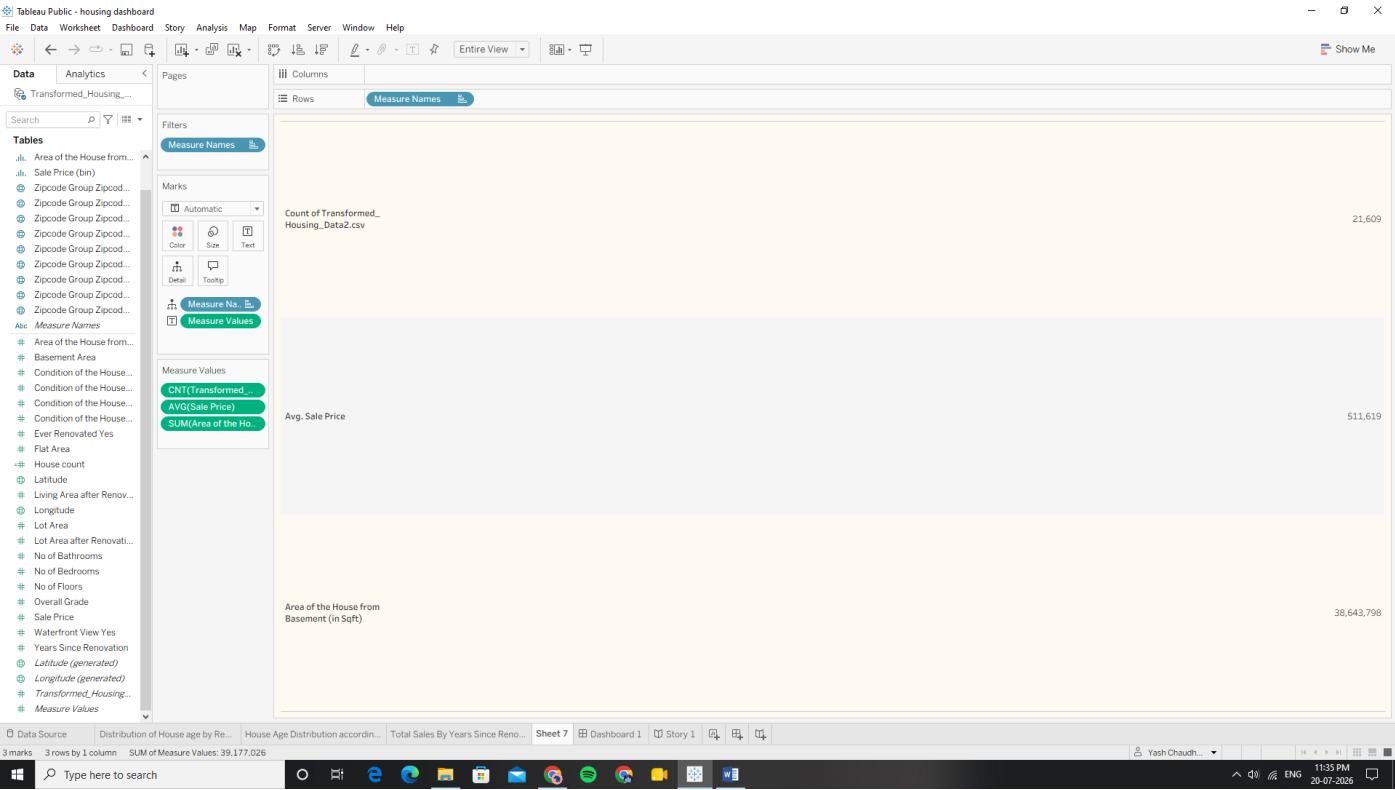


Description

The main dashboard provides an overall summary of the housing dataset by combining multiple visualizations into a single interactive interface. It allows users to analyze various housing characteristics simultaneously while applying filters to explore different segments of the data.

The dashboard serves as the central interface for understanding the housing market and provides quick access to important business insights.

Figure 7.2: Average Sale Price KPI

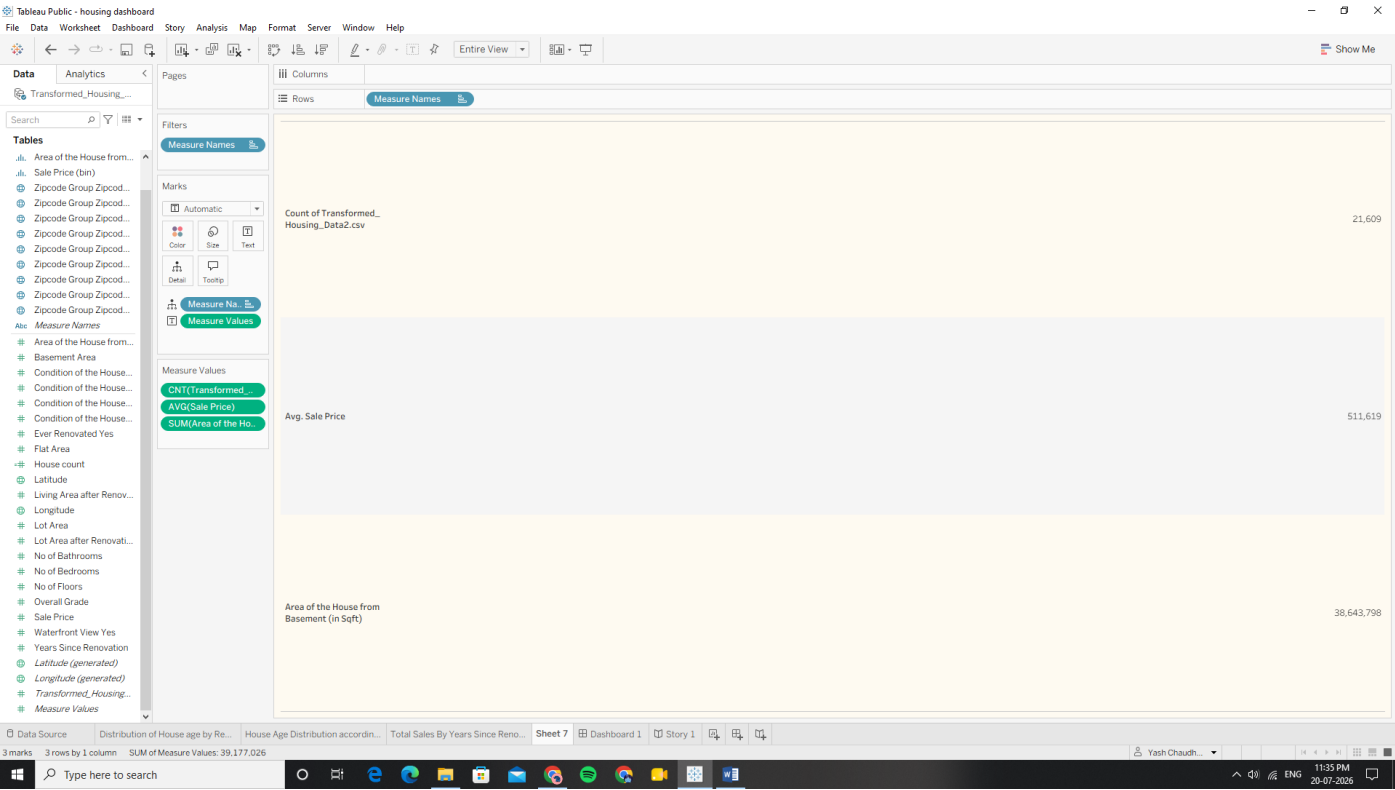


Description

The Average Sale Price KPI displays the overall average selling price of houses in the dataset. This indicator provides users with an immediate understanding of the average market value of residential properties and serves as a key performance metric throughout the dashboard.

The KPI simplifies numerical analysis by presenting a single summarized value instead of requiring users to calculate averages manually.

Figure 7.3: Basement Area Analysis

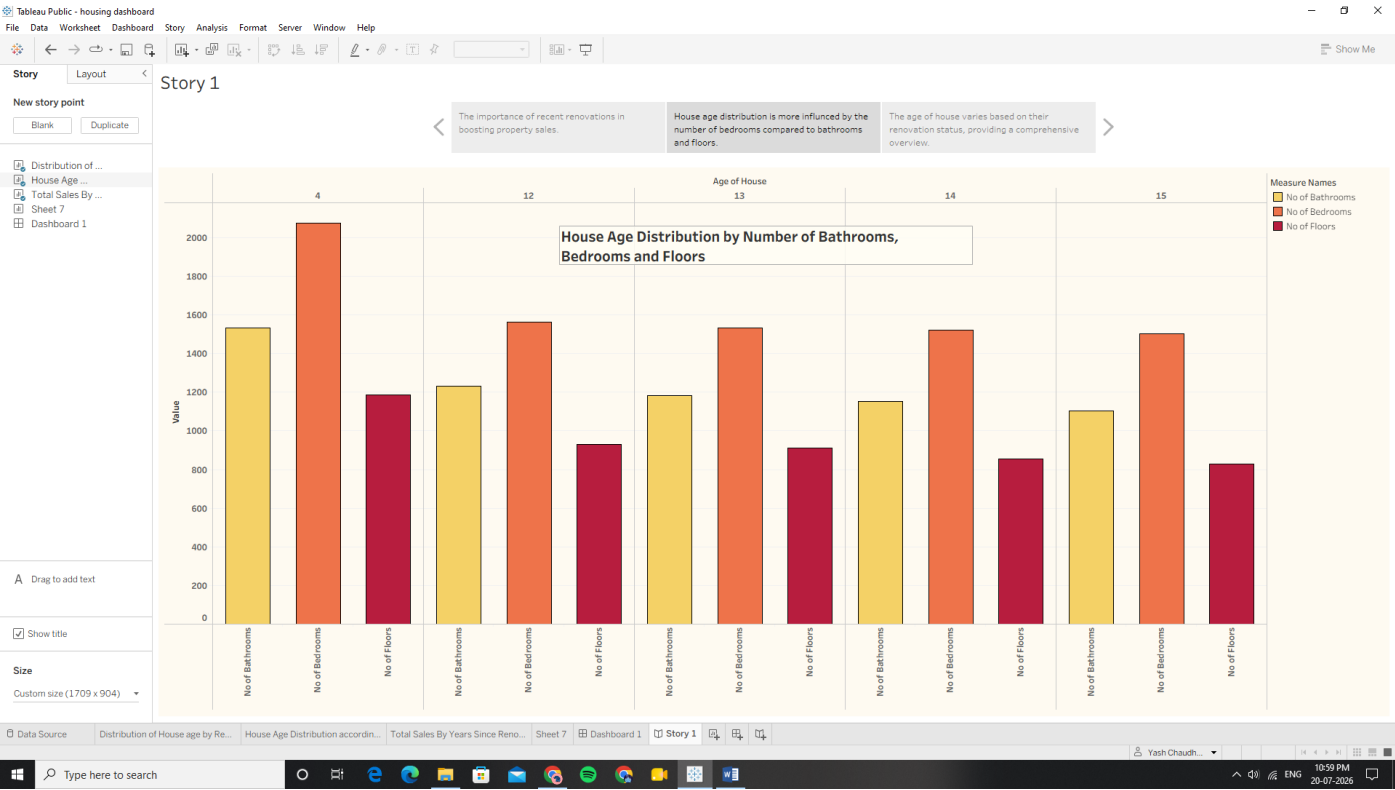


Description

This visualization analyzes the relationship between basement area and housing prices. It helps users determine whether larger basement areas contribute to higher property values.

The dashboard assists real estate analysts and property buyers in evaluating the influence of basement space on market pricing.

Figure 7.4: House Age Analysis

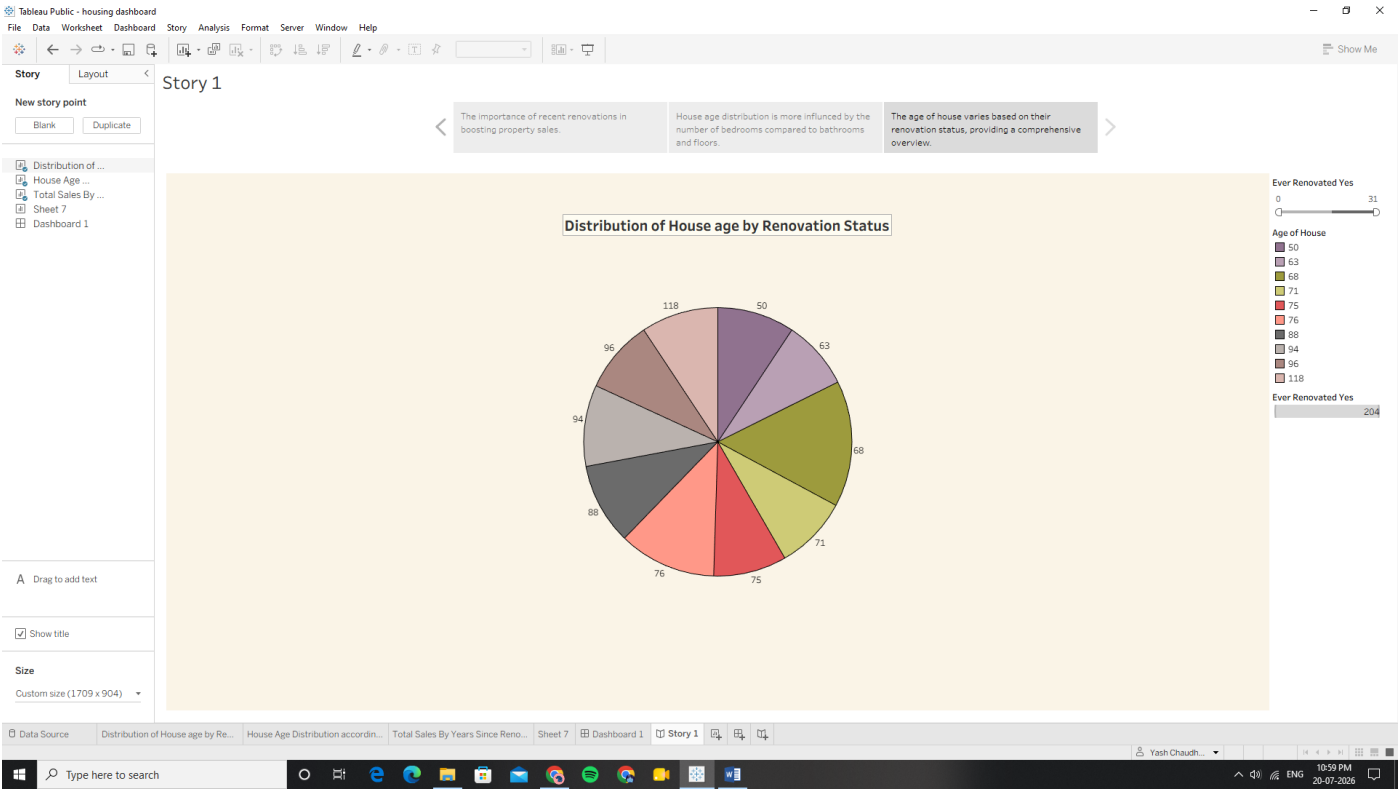


Description

The House Age Analysis dashboard illustrates the distribution of houses based on their age and allows users to observe how property age influences selling price.

This visualization provides useful information for understanding market preferences regarding newer and older residential properties.

Figure 7.5: Renovation Analysis

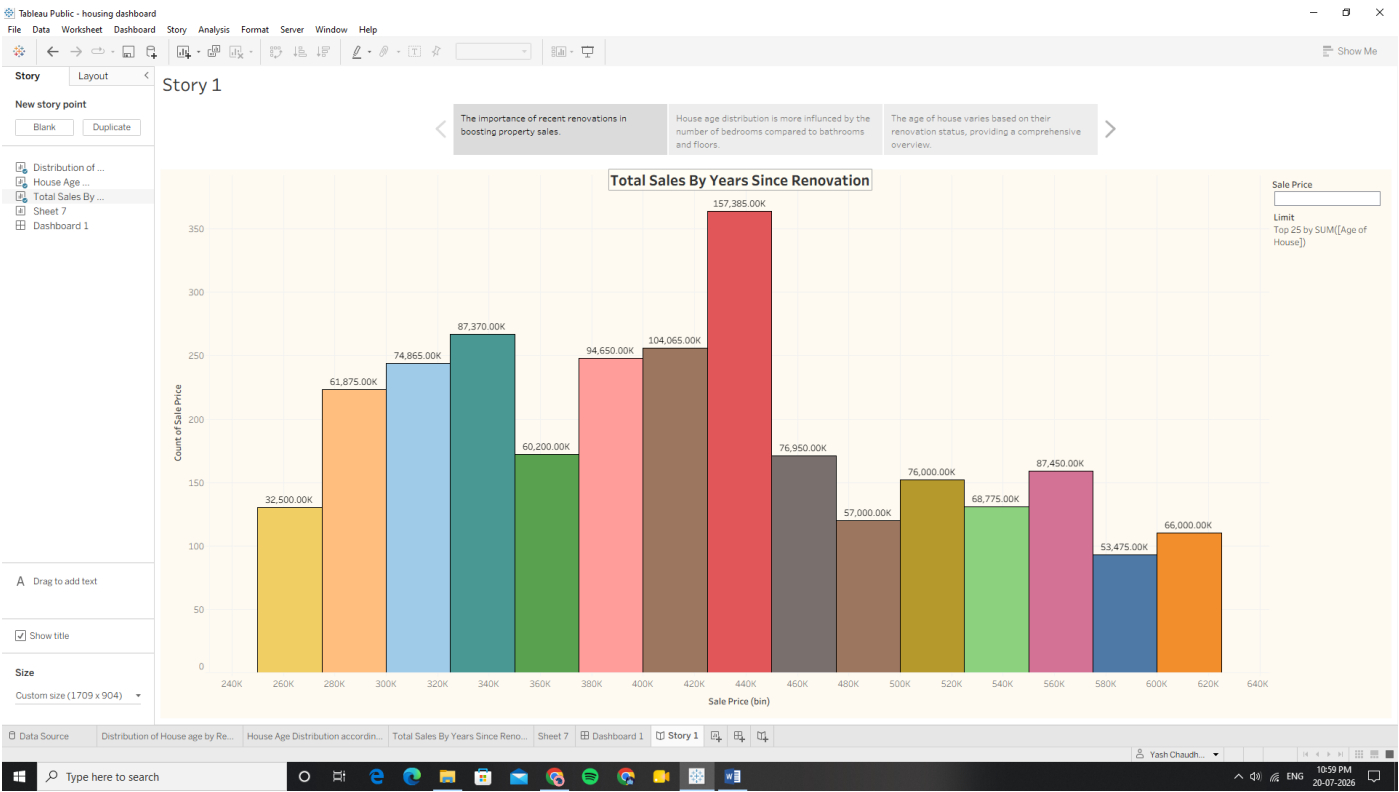


Description

The Renovation Analysis dashboard compares renovated and non-renovated houses. Users can observe differences in property prices and identify whether renovation has a significant impact on market value.

The visualization assists homeowners, investors, and real estate professionals in evaluating renovation as an investment strategy.

Figure 7.6: Sale Price Distribution



Description

The Sale Price Distribution chart presents the frequency of houses within different price ranges. It enables users to identify common pricing categories, detect outliers, and understand the overall distribution of housing prices.

This analysis provides valuable information regarding market trends and pricing behavior.

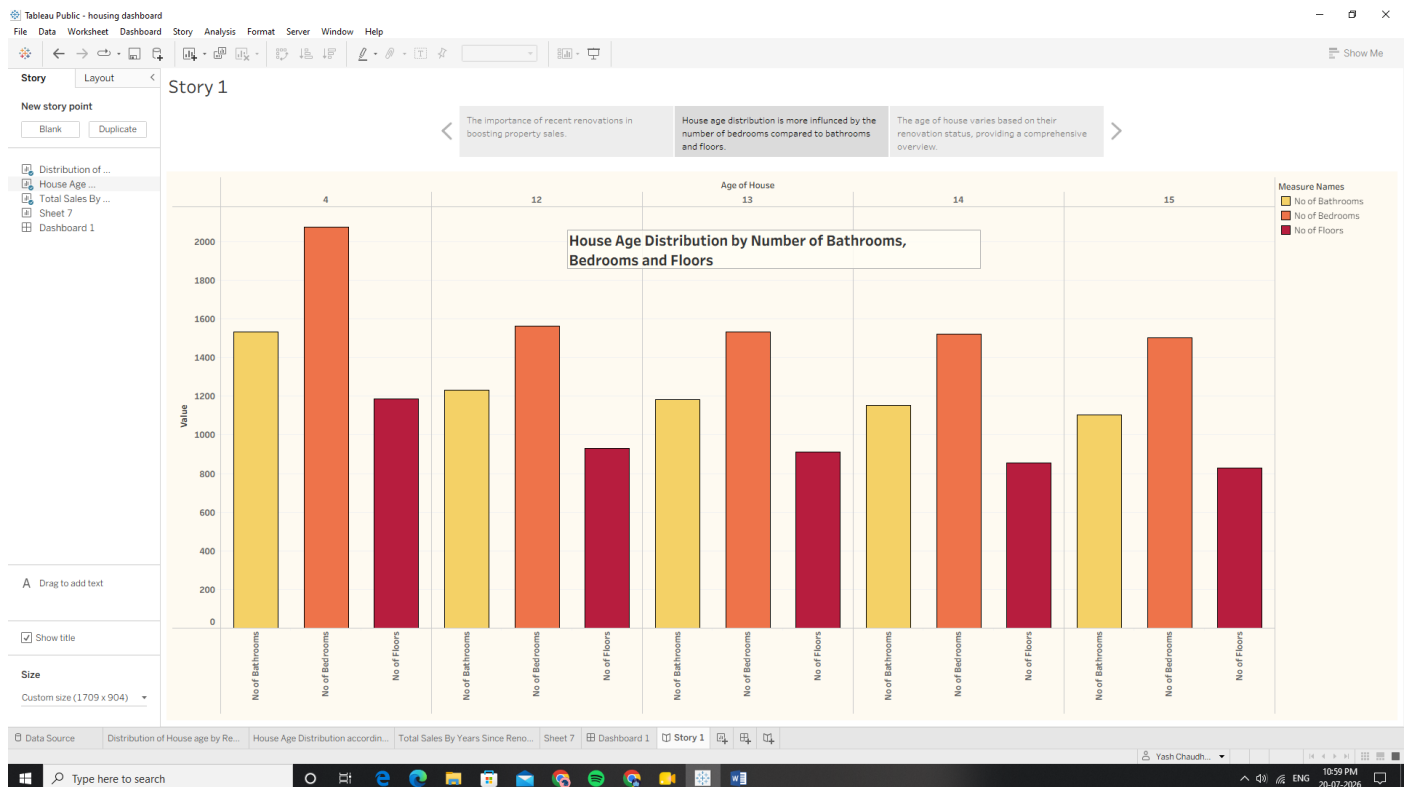
Figure 7.7: Interactive Filters

Description

Interactive filters enable users to customize dashboard analysis according to specific requirements. Users can filter information based on housing characteristics and instantly update all connected visualizations.

These filters improve dashboard usability by allowing personalized exploration of the dataset.

Figure 7.8: Tableau Story Points



Description

Story Points organize multiple dashboards into a logical sequence, allowing users to understand the complete analysis step by step. This feature improves presentation quality and helps communicate business insights effectively.

The Story Point presentation provides a structured overview of the complete housing analysis process.

Overall Findings

The analysis of the housing dataset produced several meaningful observations:

- Interactive dashboards simplified the exploration of large housing datasets.
- Housing prices were influenced by multiple property characteristics rather than a single factor.
- Renovated houses generally demonstrated higher market value compared to non-renovated properties.
- Basement area contributed positively to overall property valuation.
- Interactive filters allowed users to perform customized analysis efficiently.
- Tableau dashboards significantly reduced the time required for data interpretation compared to manual spreadsheet analysis.
- Story Points improved the presentation of analytical findings by arranging dashboards in a logical sequence.

Result Summary

The developed Tableau dashboards successfully transformed raw housing data into interactive business intelligence reports. The project demonstrated how visualization techniques improve data interpretation and

support informed decision-making in the real estate sector. The dashboards are interactive, user-friendly, and capable of providing meaningful insights to property buyers, investors, researchers, and business analysts.

The successful completion of this project confirms that Tableau is an effective tool for housing data visualization and business analytics.

8. ADVANTAGES & DISADVANTAGES

Every data analytics solution has its own strengths and limitations. The **Housing Data Analysis using Tableau** project was developed to simplify the analysis of housing market data through interactive visualizations. While the project offers several advantages, certain limitations exist due to the scope of the dataset and the technologies used.

8.1 Advantages

The major advantages of the proposed system are as follows:

- **Interactive Data Visualization:** Tableau converts raw housing data into attractive and interactive dashboards, making information easier to understand.
- **Quick Decision-Making:** Users can analyze housing prices, property features, and market trends without manually processing large datasets.
- **User-Friendly Interface:** The dashboards are simple to navigate and require minimal technical knowledge.
- **Accurate Business Insights:** Charts, KPIs, and filters help users identify important trends and relationships within the housing dataset.
- **Time Efficient:** Interactive dashboards reduce the time required to analyze housing information compared to traditional spreadsheet-based analysis.
- **Dynamic Filtering:** Users can customize analysis using filters, enabling focused exploration of specific housing characteristics.
- **Better Data Interpretation:** Visual representations improve the understanding of complex numerical data.
- **Supports Multiple Users:** The dashboard can be used by property buyers, investors, researchers, business analysts, and real estate professionals.
- **Scalable Design:** The dashboard can accommodate additional housing records with minimal modifications.
- **Improved Reporting:** Tableau Story Points provide a structured method for presenting analytical findings.

8.2 Disadvantages

Despite its advantages, the project has some limitations:

- The analysis depends on the quality and completeness of the available dataset.
- Incorrect or missing data may affect visualization accuracy.
- Tableau Desktop requires software installation and licensing.
- The dashboard analyzes historical data and does not predict future housing prices.
- Machine learning models are not included in the current implementation.
- Real-time data integration is not supported.
- Geographic analysis is limited by the available location data.
- Dashboard performance may decrease when handling extremely large datasets without optimization.

Summary

Overall, the advantages significantly outweigh the limitations. The developed dashboard provides an efficient and practical solution for housing data analysis while offering opportunities for future enhancements.

9. CONCLUSION

The **Housing Data Analysis using Tableau** project successfully achieved its objective of transforming raw housing data into meaningful business insights through interactive visualizations. The project demonstrated the practical application of Business Intelligence techniques for analyzing housing market information and presenting it in a clear, user-friendly format.

The project began with understanding the problem of analyzing large housing datasets manually. After identifying user requirements, the housing dataset was cleaned, preprocessed, and imported into Tableau Desktop. Various worksheets, dashboards, KPIs, and Story Points were developed to analyze important housing characteristics such as sale price, house age, renovation status, basement area, and other property features.

The interactive dashboards allow users to explore housing data efficiently through dynamic filters and graphical visualizations. Instead of manually examining large spreadsheets, users can quickly identify trends, compare property characteristics, and gain valuable insights that support informed decision-making.

Throughout the internship, practical knowledge of data preprocessing, dashboard development, business intelligence, and data visualization was gained. The project also strengthened problem-solving abilities and provided valuable experience in applying analytical techniques to real-world datasets.

Overall, the project demonstrates that Tableau is an effective platform for developing interactive dashboards capable of simplifying complex data analysis. The completed system provides meaningful insights for property buyers, investors, researchers, and business analysts while highlighting the importance of data-driven decision-making in the real estate sector.

10. FUTURE SCOPE

Although the current project successfully analyzes housing data, several enhancements can further improve its capabilities.

The future scope of the project includes:

- Integration of real-time housing market data.
- Development of predictive models using Machine Learning.
- Forecasting future housing prices using Artificial Intelligence.
- Addition of advanced statistical analysis features.
- Integration with cloud-based databases.
- Development of a web-based version of the dashboard.
- Mobile-friendly dashboard implementation.
- Inclusion of additional geographical analysis using GIS.
- Automatic report generation.
- Integration with SQL databases and live data connections.
- Enhanced security and user authentication features.
- Expansion of the dataset to include multiple cities and regions.
- Comparative analysis of housing markets across different locations.
- Development of personalized dashboards for different user categories.

These enhancements would increase the usefulness of the system and make it suitable for large-scale commercial applications.

11. APPENDIX

The appendix contains additional resources and references used during the development of the project.

11.1 Software Used

- Tableau Desktop
 - Microsoft Excel
 - GitHub
 - Microsoft Word
 - PDF Reader
-

11.2 Hardware Used

- Intel Core i3 (or higher)
 - Minimum 4 GB RAM
 - Windows Operating System
 - Internet Connection
 - Standard Personal Computer
-

11.3 Dataset

Dataset Name: Housing Sales Dataset

Dataset Format: CSV

Dataset Description:

The dataset contains information related to residential properties, including selling price, number of bedrooms, bathrooms, floors, living area, basement area, house age, renovation status, geographical coordinates, and other housing characteristics used for visualization and analysis.

Dataset Link:

https://storage.googleapis.com/kaggle-data-sets/5179102/8646993/compressed/Transformed_Housing_Data2.csv.zip?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=gcp-kaggle-com%40kaggle-161607.iam.gserviceaccount.com%2F20260719%2Fauto%2Fstorage%2Fgoog4_request&X-Goog-Date=20260719T180010Z&X-Goog-Expires=259200&X-Goog-SignedHeaders=host&X-Goog-Signature=81cdbfdca1a75fe3fe635843e90d6ddcc04671d2e7e968ba7ac36eb7a40a8e0ee1677e661a78f022d11b99931330e7e26de97a02fcc2e66c4693a194e34a1a2a422e2f3bdcc9b91707c50382712acba347e3d0d95a19bc2d66e4948190649a4d0ef2c60831dc1841044e6e6171c16589d718294a59f234bc61e6ad2f29199e3721a332dc9f32f0d7c02f7a12f77d4753e658730026b7f9f99e0ef1c11dd0c7c11abcf503ea032936d634dd3bfd738730330455c97e727d0757cd6262db344afe5bd3d1b719e2692dca89a2ae90607db2d8cfee9c4cdcc1cc45aaf67d8afa3ed956ac177e677ae282077f542f267d2c375dbfa519c9019b270280a9ba49cccdc4

11.4 Source Code

This project was developed using Tableau Desktop, which primarily involves visual analytics rather than traditional programming. Therefore, no programming source code was required. The Tableau Workbook (.twb/.twbx) serves as the main project file.

11.5 GitHub Repository

Repository Name:

Housing-Data-Analysis-Tableau

Repository Link:

<https://github.com/RohitonGit-Hub/Housing-Data-Analysis-Tableau.git>

11.6 Project Demonstration

Tableau Public Dashboard (if published):

https://public.tableau.com/views/housingdashboard_17844947290840/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

https://public.tableau.com/views/housingdashboard_17844947290840/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link